

Curriculum Vitae

Muskaan Irfan

Data Scientist | Mathematician | Top Coder | National Badminton Player |
Int'l Hackathon Judge | Merit Scholarship Holder
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PROFILE

- Invited by the Government of Japan as a Visiting Researcher under the JST Sakura Science and Technology Program at University of Fukui.
- Featured by a Japanese magazine for contribution to research.
- Winner of international competitions, including Stanford, Harvard, and MIT.
- Equally adept in the classroom and on the badminton court.
- Merit scholarship holder from elementary school till MS.
- National badminton player and international hackathon enthusiast.
- Well-versed in taking failures as lessons and keep crawling till it is done.
- Comfortable working in cross-cultural, cross-time-zone teams under tight deadlines.
- International Hackathon participant, team lead, judge, and mentor.
- If someone can dream it, I can code it.

ACHIEVEMENTS

- Visiting Researcher for JST Sakura Science and Technology Program at University of Fukui, **Japan,** for 3 weeks in 2026 (Link)
- Winner of CS50x Puzzle Day 2024, 2025, 2026, organized by **Harvard University** (Link)
- Featured by Tannan FM Magazine (Fukui, **Japan**) for contributions to international research and collaborative innovation 2026 (Link)
- Recognized with Grants Excellence and Operations Award for contributions as ML/AI researcher by **Virufy, USA**, 2026 (Link)
- Section Leader/TA for **Stanford's CIP** 2025 (Link)
- Secured 114th position globally in the **MIT Informatics Tournament 2025,** competing among 530+ teams (Link)
- Ranked **Top 29% globally** out of 15000+ in **Meta Hacker Cup 2025** Round 1 and **6th in Pakistan (Top 3% globally)** in the Practice Round (Link, Link)
- Ranked **2nd** among female coders in Pakistan (globally ranked 12th) in the **Advent of Code 2025** leaderboard (Link)
- Achieved 102nd global rank out of 860+ teams in the **UC Berkeley Coding Competition 2024** (Link)
- Runner-up in the International Youth Math Challenge 2021
- Achieved Appreciation Award 2022 from Fixit (An NGO dedicated to civic welfare and education) for voluntary contributions to their student mentorship program

EDUCATION

MS Data Science Sep'23 - Sep'25
Air University Pakistan
CGPA: 3.68/4
Relevant Coursework: Machine Learning, BigData Analytics, Data Visualization, Deep Learning, Statistical and Mathematical Methods for Data Analysis, Computer Vision, NLP.

Thesis: *Advancing Human Activity Recognition Through Deep Learning and Sensor Fusion* (PDF)

- Developed an LSTM-based deep learning model for multivariate sensor time-series classification.
- Achieved 97% testing accuracy through advanced preprocessing, feature extraction, and sensor-fusion strategies.
- Designed an open HAR dataset and implemented robust evaluation metrics to validate model performance.
- Expected Paper Submission date: June 2026

Awards:
- Won merit scholarship
- Secured 1st position in both *Statistical & Mathematical Methods for Data Analysis* and *Machine Learning* across five Creative Technologies departments.

BS Mathematics Oct'18 - Jun'22
Quaid-i-Azam University, Pakistan
CGPA: 3.2/4
Awards: Won merit scholarship

PUBLICATIONS

1. Sulthan, N., Daryanani, A., Dharmi, S., May, G., **Muskaan, M.**, Sreekumar, G., Saito, R., Amer, S., Ahmed, I., & Aslam, A. (2025). "Pneumonia and COPD in Middle Eastern Older Adults: Current Burden and Future Opportunities", **Published in** *International Journal of Applied Technology in Medical Sciences*, 4(2), 4-25. https://doi.org/10.54878/44jjb542

2. A. Khanzada, **Muskaan,** L. MacCabe, "From Volunteer to Vocation: The Career Impact of Skill and Network Development in a Global Health Technology Nonprofit", **Submitted** to *Journal of International Training and Development* in April 2026. https://doi.org/10.5281/zenodo.19883684

3. A. Khanzada, L. Atlas, N. Rasmussen, F. Khan, **Muskaan,** "Acoustic Phenotyping of Coughs Predicts COVID-19 Screening Accuracy", **Submitted** to *Journal of Artificial Intelligence Research* in May 2026 (PDF*)*

SKILLS

Programming Languages: Python, C++, R
Software & Tools: MATLAB, VS Code, Google Colab, Jupyter Notebook, Spider, GitHub, Tableau, Power BI, WEKA, Overleaf, LaTeX
Framework/ Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, PyTorch, SciPy, Keras, NLTK, Transformers(Hugging Face)
Databases & BigData Technologies: PostgreSQL, PostGIS, PySpark, Apache Hadoop, Apache Hive
Operating Systems: Windows

WORK EXPERIENCE

AI/ML Researcher Sep'26 - Present
Sakura Science Exchange Program
Hosted by University of Fukui, Japan

- Contributed to AI-driven tourism research using the Distributed Human Data Engine (DHDE), integrating AI-camera visitor counts, Japan Meteorological Agency (JMA) weather observations, route-search intent signals, and tourism survey data to analyze international visitor demand patterns in Fukui, Japan.

- Worked on data-driven tourism intelligence focused on visitor-demand forecasting, statistical modeling, and spatial analysis to investigate weather-related tourism barriers and identify opportunities for regional tourism development.

  Project Link: https://dhde-tourism-ai.github.io/dhde-app/

Mathematics Lecturer Feb'23 - Aug'23
IBA Public School & College Sukkur, Pakistan

- Taught Mathematics, developed curriculum and lesson plans.
- Attended Maths conferences, took advanced Maths courses, and participated in workshops.

Research Intern Aug'22 - Sep'22
National Centre Of Physics (NCP), Pakistan

- Conducted research on **Special Relativity and Cosmology**, using **MATLAB** and **Python** for simulations and modeling, gaining insights into relativistic phenomena.
- Analyzed cosmic microwave background data with CAMB and CLASS to study early universe dynamics, and utilized the Einstein Toolkit for numerical relativity simulations.

VOLUNTEER EXPERIENCE

ML/AI Research Intern June'25 - Present
Virufy (Link) Remote - California, USA

- Served as second author on an ongoing research paper, led quantitative and statistical analysis. Contributed to methodology design and results interpretation.
- Actively contributing to healthcare and AI research papers, supporting systematic literature reviews, analysis, and technical writing of manuscript development.

Recognized with **Grants Excellence and Operations Award** for outstanding contributions: Link

STEM Program Manager Feb'24 - Present
iCode Guru (Link) Remote - San Jose, CA, USA

- Organized and moderated weekly tech webinars featuring Silicon Valley, California, professional engineers.
- Taught Python, Data Structures and Algorithms and Maths courses to underprivileged students in their local language.
- Presented LeetCode problems and responded to programming queries. Guided students to participate in International Hackathons and Coding Competitions.

Section Leader/ Teaching Assistant (Link) Apr'25 - May'25
Stanford University Remote - Palo Alto, CA, USA

- Taught international students Python using Karel the Robot at Stanford.
- Helped peer mentors discuss new teaching methodologies.

Tech Mentor & Judge (Link) Aug'24 - Present
GenAI Hackathon Pak-Angels, San Jose, CA, USA

- Delivered a webinar to 500+ participants, covering hackathon strategies, project ideation, teamwork, and effective pitching techniques.
- Guided teams in developing Gen-AI applications, including RAG models and chatbots, ensuring smooth project execution and deployment.
- Evaluated 100+ projects and conducted one-on-one mentorship sessions, addressing technical queries, refining ideas, and ensuring fair recognition of outstanding work.

ACADEMIC SERVICE & CONFERENCE CONTRIBUTIONS

- **Program Committee Member, IEEE MedAI 2026:** Selected to serve on the program committee, contributing to the peer-review process for research in medical AI. (Link)
- **Research Paper Reviewer, NeurIPS & Conference Speaker:** Reviewed 10+ research papers across conferences and NeurIPS workshops, and contributed as a speaker at academic and research events. (Link)

INTERNATIONAL HACKATHONS

Cohere Hackathon - LabLab.ai (Presentation) (Link) Jan'23
Team Size: 4 | Muskaan I.(Team Lead, Pakistan)**,** Leandra P.(USA), Maha Z.(Pakistan), Ibtisam M.(India)

My Role: Project Manager & Idea Lead
Tech Stacks: Python, Cohere API (LLMs), OpenCV, NLP, Computer Vision, Sentiment Analysis

- Developed an AI Job Interview Assistant using the Cohere API, enabling candidates to record responses and analyze their verbal answers and facial expressions.
- Enhanced candidate experience and employer brand by building confidence, saving time, and streamlining the interview process.

OpenAI Hackathon - LabLab.ai (Presentation) (Link) Dec'22
Team Size: 6 | Muskaan I.(Pakistan), Nargiz S.(Team Lead, Georgia), Nik Chein(Taiwan), Ayesha A.(UAE), Hafsa N.(Pakistan), Nafia A.(Pakistan)

My Role: AI Product Contributor & Developer
Tech Stack: Node.js, Python, OpenAI API (LLMs), Google Books API, Recommender Systems

- Developed the Bibliotopia App, an AI-powered bookshop leveraging GPT-3 to provide personalized book recommendations and promote literacy initiatives.
- Collaborated with humanitarian organizations, Google Books, government agencies, and libraries to drive literacy programs, managing product development, AI engineering, and customer service.

CONFERENCES, WORKSHOPS, AND WEBINARS

- Conducted 6-hour workshops on *Python Data Visualization for Mathematicians*, focusing on Matplotlib, Seaborn, and practical charting techniques at iCodeGuru (Link)
- Delivered a webinar, *Beginner's Guide to Join Hackathons* at iCodeGuru & Aspire-Pak (Link)
- Organized a Session on *Unity in Diversity and Pluralism by Centre for Peace Education Pakistan* (Link)
- Organized a workshop on *Startup for Applied Researchers in Pakistan* (Link)

HOBBIES/INTERESTS

- **Hackathons:** Actively participate in international hackathons, collaborating with diverse teams to create innovative projects.
- **Writing:** Share insights on technology, data science, and algorithms through articles on LinkedIn, engaging a wide audience.
- **Badminton:** Competed at the National Olympics, proudly representing Quaid-i-Azam University, demonstrating high-level athletic performance, discipline, and team spirit.
- **Artistic Skills:** Artistic pursuits include painting, calligraphy, portrait sketching, and creating custom artwork. (Link)